

Simulated Ensemble Performance Partners: Exploring the Integration of Generative AI and Virtual Reality for Collaborative Rehearsals

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Abstract

Ensemble musical performances demand both technical precision—such as exact fingerings, intonation, and tempo—as well as higher-order skills such as active listening, real-time responsiveness, and emotional expression. This study explored the ways that generative AI and VR could be used to facilitate this complex interplay of cognitive and expressive demands. We conducted a needs-finding pilot study with four advanced viola students who rehearsed in a gallery-style virtual concert hall accompanied by a smartphone-based AI partner. Post-rehearsal interviews underwent thematic analysis to identify key system requirements for simulating an effective ensemble partner. Three core needs emerged: (1) faithful interpretation of score arrangements to support expressive nuance, (2) adaptive sensory feedback to enhance VR immersion, and (3) pre-rehearsal exchange of emotional intent. The insights gained in this study could guide the development of AI/VR systems aimed at augmenting ensemble practice, supporting iterative rehearsal in immersive environments that complement human musicianship.

CCS Concepts

• **Human-centered computing** → **Empirical studies in collaborative and social computing**; **Empirical studies in HCI**.

Keywords

Ensemble Music Performance, Collaborative Rehearsal, Generative AI, Virtual Reality, Technology Probe.

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1 Introduction

Ensemble musical performance is a creative endeavor that demands more than technical mastery, such as precise fingerings, flawless intonation, and a steady tempo [6]. Its depth lies in active listening and mutual responsiveness, as players negotiate phrasing and dynamics through visual cues and subtle nonverbal indicators—a shared breath or subtle gesture—to stay synchronized and blend their voices into a unified sound. Beyond these player-to-player interactions, true ensemble playing requires the projection of emotional content while responding to the energy of the audience, creating a living dialogue between performers and listeners.

This study sought to identify the foundational requirements for the integration of generative artificial intelligence (AI) and Virtual Reality (VR) to simulate an ensemble-partner for rehearsal purposes. Among various instruments, the string and keyboard instruments form the backbone of many chamber and orchestral ensembles, with stringed instruments providing rich harmonies and melodic lines, and keyboards supplying broad-range harmonic support. Professional practitioners of string and keyboard instruments are expected to navigate a wide spectrum of styles—from solo repertoire to chamber music to full orchestral works [6]. [8, 16]. Therefore, a system integration of generative artificial intelligence (AI) and Virtual Reality (VR) to simulate an ensemble-partner for rehearsal purposes could be used as a complement to human collaborators, while allowing limitless experimentation—allowing musicians to refine technical passages, explore expressive choices, and rehearse in a richly immersive, AI-driven environment.

Most of the AI systems developed for the refinement of musical skills focus on solo-practice—such as reminders or suggestions for the performance of compositions by a single musician [9]. Some musicians avail themselves of pre-recorded tracks, such as YouTube videos, or VR simulations to emulate audition settings or large venues with a live audience [4, 7]. Fanger et al. [4] developed a mixed-reality system that gradually immerses the performer in virtual audiences to build resilience to distractions and performance pressures. Lecamwasam et al. [7] demonstrated the use of real-time visual feedback to help musicians regulate their emotions and maintain equanimity. Maezawa and Yamamoto [10] created a machine-learning ensemble system that uses audiovisual cues to synchronize precisely with human performers, thereby enabling performers to deliver expressive intent with poise and confidence. Other innovations include MyPianist, which generates real-time accompaniment based on live tempo tracking [12], and an improvising AI drummer that visualizes its confidence using emoticons—enhancing engagement through transparent physiological and musical signaling [11]. However, these systems often become desynchronized when performers introduce dynamic or expressive variations during rehearsals. Moreover, none of these systems support collaborative designs capable of recreating the experience of rehearsing with a live ensemble.

This needs-finding pilot study involved a technology-probe session with four advanced students of music performing a collaborative performance with a smartphone-based AI accompanist in an immersive VR environment complete with a virtual concert hall, 3D audience avatars, and ambient sounds [12]. Post-rehearsal interviews explored the musical backgrounds of the students as well as the challenges they face in rehearsing ensemble compositions, and their perceptions of AI/VR-based rehearsal tools. Thematic analysis of interview transcripts revealed three essential characteristics for an effective ensemble partner: (1) faithful interpretation of score arrangements to support expressive nuance, (2) adaptive sensory feedback to enhance immersion during VR practice, and (3) pre-rehearsal exchange of emotional intent. The insights gained in this study could guide the development of AI/VR systems aimed at augmenting ensemble practice, supporting iterative rehearsal in immersive environments that complement human musicianship.

2 Pilot Study

Needs analysis in the current study combined hands-on technology-probe sessions with one-on-one interviews. Recruitment was conducted via snowball sampling through a social network of one of the co-authors—a viola major and faculty member in the Department of Music at National Tsing Hua University. The participants included three females and one male (mean age = 21.5 years, SD = 2.65, range 19–25). All participants cited the viola as their primary instrument, but two (M1, M3) also play the piano. Each participant had at least 5-years of experience playing their main instrument and had appeared in more than ten performances.

A review of the literature confirmed generative AI—powered by real-time performance data from wearable sensors—can achieve highly engaging human–computer musical interactions [11, 14]. Building on these insights, the proposed technology probe delivered a generative AI simulation via a Meta Oculus 3 VR headset.

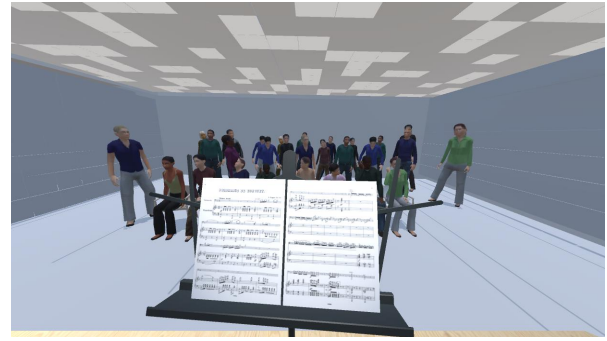


Figure 1: In the simulation of the ensemble performance, the sheet music rests on a stand in front of 3D audience avatars.

Sessions were conducted in two phases: Preparation (2 minutes) and Rehearsal (3 minutes). During the preparation phase, the MyPianist app (running on a mobile device) recorded the participant’s performance and compared it to its default accompaniment. Based on real-time analysis of the participant’s playing, the app dynamically adapted the timing and expression of its accompaniment to align with the live performance. This real-time analysis enabled the system to learn the participant’s stylistic nuances, particularly in phrasing and dynamics. In the subsequent rehearsal phase, the AI accompanist leveraged these learned insights to continuously adapt its accompaniment in real time, aligning with the participant’s evolving performance.

Consistent with the technology-probe methodology [5], the system was designed as a functional prototype rather than a polished, production-ready platform. Its primary aim was to provoke authentic user reactions and generate feedback to guide future design iterations. Researchers began by outlining the objectives of the pilot study and providing detailed explanations of the procedures. They then assisted the participants with donning the VR headset. Once inside the simulation, participants were transported to a spacious, gallery-style concert hall [3] populated with 3D audience avatars from the Audience On Stage asset project [13] and ambient sound clips of crowd chatter, coughs, and chair creaks (downloaded from free online sharing platform [2]). Within this realistic environment, they performed Stamitz’s Viola Concerto in D major [15], accompanied by the MyPianist app running on a smartphone linked to external speakers [12].

Following the rehearsal phase, each participant took part in a semi-structured interview covering three core areas: (1) their journey to majoring in their primary instrument and their musical learning process; (2) challenges they currently face when rehearsing with human ensemble partners; and (3) how generative AI and/or VR could support or enhance ensemble rehearsals. Each interview lasted between 90 and 120 minutes.

All interviews were audio-recorded and transcribed for thematic analysis [1]. During the first round of open coding, we identified segments describing rehearsal challenges and the ways AI and VR could address them. The research team then met to review and refine the initial codes, discussing the meaning and relevance of key quotations. A second round of axial coding was conducted to organize these refined codes into broader categories, revealing

emergent themes and identifying specific design enhancements needed to support the goals articulated by participants.

2.1 Requiring the AI to incorporate musical score arrangements to create more heartfelt accompaniment

All of the participants reported a long-standing reliance on pre-recorded concert performances from platforms like YouTube. Despite its convenience, these static tracks are unresponsive—offering no adjustment to the player’s phrasing, tempo, or expressive choices. Participants found that, unlike fixed recordings, the AI accompanist could adapt its tempo in real time. For example, M04 commented *"During the two-minute preparation phase, the accompanist lagged behind during the rapid arpeggios and couldn't maintain my tempo. However, after that run-through, the system learned my phrasing and dynamics. When we replayed the rapid arpeggios in the subsequent rehearsal session, there was no delay—the accompaniment tracked my playing more accurately. Remember that I couldn't see my instrument and I was unfamiliar with the concerto, so I missed a few notes. The system misinterpreted those omissions as a tempo increase and adjusted its tempo to match what it perceived as my new speed."*

However, some users noted that the system occasionally misinterpreted expressive timing. Arpeggios, which involve rolling chord tones in rapid succession, can feature stretched or compressed durations for expressive effect. In some cases, the AI misread these expressive liberties as timing inconsistencies. For example, M02 commented *"When the chord is played as an arpeggio, the notes are split up. If I play the first two notes slightly early so that the final note lands exactly on the downbeat, the system tends to misinterpret my timing. In my simulation, the AI thought that the final note on the downbeat was actually the first note of the arpeggio and was still 'waiting' for the other two notes to arrive together. It interpreted that as me lagging—even though my tempo remained constant—so the accompanist slowed down to match what it thought was my new tempo."*

Participants expressed hope that future systems could not only follow their phrasing but also respond with greater expressive nuance. For example, M02 commented *"This concerto features clearly defined tempos and expressive markings, characteristic of works from the Classical period. That made it easier for the AI to 'read' my performance—dynamics and timing stay within a conventional classical framework—making it easier for the accompaniment to follow. But other styles, such as modern and Romantic works, often involve more tempo rubato, unexpected pauses, or nuances that are not outlined in the score. This greater interpretive freedom could confuse the current AI system. I would expect that pieces with more dramatic tempo shifts or wider expressive flexibility could become messy—but they might also present exciting challenges for the system to tackle."*

2.2 Incorporating Evocative Sensory Elements to Enhance Presence During Ensemble Practice

All participants envisioned using VR to recreate realistic performance settings that simulate the psychological and sensory conditions of live ensemble work. For example, M01 commented *"I*

thought of recreating an audition setting—imagine an orchestra audition where a curtain hangs before you and the judges sit unseen behind it. Simulating an environment would make me feel like I'm concealed behind a curtain, with all the accompanying tension."

Several participants expressed surprise at the subtle audio and visual details embedded in the virtual environment. They described how these immersive elements sharpened their focus and helped build the confidence needed to remain composed during live performances. For example, M02 commented *"Once I put on the helmet, I was astonished by the virtual audience avatars—they really transported me onto the stage. The VR-rendered sheet music surprised me with its crisp clarity and readability. As you study the sheet music in VR, you'll sense people shifting around and even hear their voices or coughs, adding to the realism. Practicing in this fully immersive environment helps you build confidence so that when it's time for a real performance, you won't feel as nervous."*

Others proposed further enhancements to the lighting design—such as a focused spotlight on the performer—to amplify the drama and deepen the emotional impact of the virtual ensemble performance. M04 commented *"It would be great to enhance the venue lighting—adding a spotlight, dynamic stage lights, or ambient mood lighting. I believe these effects would heighten my sense of tension. For example, shining a spotlight on me would create a more formal, on-stage ambiance."*

Participants also drew parallels to chamber music rehearsals, emphasizing the importance of incorporating visual cues—such as eye contact, a shared breath, or a preparatory gesture—to coordinate with ensemble members. They felt that these nonverbal elements should be included in future VR environments to improve realism and synchrony. M04 commented *"Chamber music relies on visual cues—sometimes a shared breath or a preparatory gesture. For instance, you need eye contact or a subtle nod to start together. Whether you place a virtual piano beside me or not, having that visual reference makes it feel far more authentic, as if you're truly sharing the stage."*

2.3 Exploring Collaborative Dialogue Between Musicians and AI Companions to Optimize Ensemble Rehearsal Dynamics

Even as the accompanist subtly synchronized its tempo with participants in the second rehearsal phase, many found themselves unconsciously adjusting their own timing to match it—prompting uncertainty over who was truly leading. This ambiguity occasionally caused hesitation or performance errors. For example, M03 commented *"After listening to both recordings, I noticed that in the first trial [during the preparation phase] I was negotiating the tempo with the AI. In the second trial [during the rehearsal phase], the accompanist didn't start any faster—in fact, it maintained a steadier tempo—and it seems the system adapted its playing based on my first-round performance by analyzing my timing and saving it for next time. Even so, in both trials I found myself constantly trying to matching its tempo and second-guessing, 'How fast should I really go?' I was never quite sure who was following whom, so I improvised in the moment."*

Several participants reported that the accompanist struggled to maintain a steady tempo in certain passages. During instrumental interludes or rhythmically demanding sections, the system would

lag or surge ahead, leading performers to wonder whether they should follow the AI or stay true to their intended pacing. For example, M01 commented *"There were moments when the accompanist couldn't keep up. During some interludes—after I'd finished my part and only the interlude remained—I noticed the system slowing down. Then I'd wonder to myself, 'since we're playing together, should I slow down when the AI slows down, or should I stick to my original tempo?' Likewise, there are passages where the rhythm and chords should align perfectly, but the AI still seemed to lag behind or rush ahead."*

This led participants to express a need for clearly defined dynamics and expressive contours in advance—something commonly discussed in live ensemble rehearsals. Without predefined interpretive guidance, the accompaniment felt flat or emotionally monotonous. For example, M04 commented *"I found the performance of the AI was somewhat flat throughout. In this piece, many melodic phrases span two bars, so we create contrast by alternating between soft and loud passages. It is normal to coordinate these dynamics—deciding together which sections the accompaniment should play softly and which it should emphasize."*

Participants emphasized the importance of pre-communicating tempo shifts and expressive intentions to ensure smooth coordination, especially in passages that rely on clear phrasing or call-and-response structures. For example, M04 commented *"Although this piece doesn't include non-overlapping exchanges, it is sometimes necessary to perform a call-and-response with the piano, where the pianist plays a phrase, and I immediately repeat it so our parts never overlap. This alternating exchange adds an extra layer of complexity. [...] Also, everyone has personal habits about how to handle the end of a phrase. Most people use ritardando there, while others prefer to keep the original tempo. If you plan to slow down, discuss it with your partner in advance, saying something like, 'I'll ease into this slower tempo,' and specify exactly how much I'll decelerate."*

Given these challenges, participants made it clear that future systems should include a built-in communication interface for pre-defining the tempo and expressive changes. Without such coordination, the performance would feel mechanical. Allowing performers to preset dynamics and phrasing would greatly enhance its emotional richness. As M02 commented *"I would like to discuss with the system how it intends to express emotion beforehand. Any discussion would make the performance better. As for emotion, right now I feel the only control we have is simple commands like 'hold this note for an extra second' or 'increase volume here.' Today the accompaniment's dynamics and durations were all uniform. If in the future it could be customizable—so each performer could design these details—it would greatly enhance the emotional expressivity."*

3 Preliminary Discussion

Building on our pilot findings, future research should focus on integrating generative AI and VR to develop an ensemble performance partner capable of conveying both musical nuance and emotional depth. First, it is crucial that the system faithfully interpret score arrangements—honoring phrasing, dynamics, and structural contrasts—to generate expressive, heartfelt accompaniment. Evocative sensory elements—such as lighting cues, ambient audience sounds, haptic feedback—should also be added to simulate the rehearsal environment and deepen immersion. Finally, there is a pressing

need for collaborative dialogue between the musicians for the sharing of emotional intentions, call-and-response cues, and tempo adjustments. Whether these decisions are made prior to or during rehearsal, such interactions can significantly enhance score fidelity, refine technical precision, and foster interpretive confidence in preparation for live performances. This pilot study focused exclusively on viola players. However, it is important to recognize that score interpretation and accompanist requirements vary across instruments. To broaden the applicability of our findings, our future work will expand participant recruitment to include violinists, cellists, and pianists. Subsequent design efforts will aim to identify and incorporate features that address the shared and instrument-specific needs of both string and keyboard musicians.

Some participants also reflected on the optimal stage for introducing an AI ensemble partner. M02 suggested that the system would be most beneficial midway through preparation—after the musician has already mastered the notes, intonation, and tempo. Introducing the AI too early, while still learning the part, might be more distracting than helpful. All participants agreed that the system should be used only after basic fingerings are secure, noting that the VR headset can hinder visibility of the instrument and distract from reading the score. Once the performer is comfortable with the piece, however, the AI accompanist can be used to explore interpretive possibilities and rehearse different accompaniment patterns. In this way, the tool complements human collaboration while enabling novel forms of ensemble rehearsal. Nonetheless, M01 cautioned that each performance venue has its own unique acoustic properties. For example, one place may have a pronounced echo, while places like recording practice studios are well dampened. Participants emphasized that adapting one's sound to the venue is a crucial part of final rehearsal. Whether it occurs on performance day or in prior on-site sessions, in-person adjustment to room acoustics remains essential. As such, the AI/VR system is best suited as a mid-stage rehearsal aid—offering added flexibility for refining accompaniment when human collaborators are unavailable or when repeated experimentation is needed.

4 Conclusion

We are currently collaborating closely with a music expert—one of our co-authors and a viola specialist on the Music Department faculty—to refine both our technology-probe design and interview protocol. Additional participants will be recruited through referrals from this expert. The insights gained from this pilot study will inform future development of generative AI and virtual reality systems capable of simulating ensemble partners.

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